COUNTRY DEVELOPMENT CLUSTERING USING AGGLOMERATIVE CLUSTERING

MINI PROJECT REPORT

submitted by

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1. Introduction

Online retail platforms generate enormous volumes of transactional data every day. Each purchase event contains rich information about customer preferences, product demand, and buying behaviour. Despite this wealth of data, many retailers struggle to translate raw transactional records into actionable business intelligence.

This project addresses that challenge by applying Market Basket Analysis using the Apriori Algorithm to an online retail dataset. The Apriori algorithm is one of the most widely used approaches in association rule mining, capable of discovering frequent itemsets and generating meaningful product association rules based on measurable metrics such as Support, Confidence, and Lift.

1.1 Problem Statement

Online retailers face recurring challenges in leveraging transactional data: identifying products customers frequently purchase together, understanding hidden purchasing patterns, designing effective recommendation systems, and optimizing cross-selling strategies. This project uses association rule mining to systematically address these challenges by extracting statistically significant product combinations from transaction records.

1.2 Project Objectives

1. Import and preprocess the online retail transaction dataset.
2. Clean the dataset by handling missing values and irrelevant data.
3. Convert the dataset into a transactional format suitable for association rule mining.
4. Perform exploratory data analysis to understand item frequency and distribution.
5. Apply the Apriori algorithm to generate frequent itemsets.
6. Generate association rules from the frequent itemsets.
7. Evaluate rules using Support, Confidence, and Lift metrics.
8. Visualize the association rules using graphs and rule plots.
9. Identify strong product combinations frequently purchased together.
10. Provide data-driven recommendations for improving online retail sales strategies.

2. Background and Theory

2.1 Market Basket Analysis

Market Basket Analysis (MBA) is a data mining technique used to understand the purchase behaviour of customers by finding associations between products bought in the same transaction. It is widely used in retail and e-commerce industries to design promotional bundles, arrange store layouts, and generate product recommendations.

2.2 The Apriori Algorithm

The Apriori algorithm, proposed by Agrawal and Srikant in 1994, is the foundational algorithm for association rule mining. It uses a breadth-first search strategy to identify frequent itemsets and then generate rules from them. The algorithm is built on the Apriori property: any subset of a frequent itemset must also be frequent, which is used to prune the search space efficiently.

2.3 Key Metrics

Metric	Formula	Interpretation
Support	P(A and B)	Proportion of transactions containing both items A and B
Confidence	P(B|A) = P(A and B) / P(A)	Probability of buying B given that A was purchased
Lift	Confidence / P(B)	How much more likely B is bought with A vs. independently. Lift > 1 = genuine positive association

2.4 Parameter Selection

For this project, the Apriori algorithm was configured with minimum support 0.01, minimum confidence 0.50, and minimum rule length 2. Rules are sorted by Lift to surface the strongest and most practically significant associations.

3. Dataset Description

3.1 Source

The dataset used for this project is an online retail transaction record from a UK-based e-commerce platform covering purchases made between 2010 and 2011. The raw file is loaded as OnlineRetail.csv from the UCI Machine Learning Repository.

3.2 Dataset Statistics

Property	Value
Source	OnlineRetail.csv (UCI Machine Learning Repository)
Platform	UK-based online gift and home decor retailer
Format	Transactional CSV with invoice-level records
Algorithm	Apriori (arules package in R)
Parameters	supp = 0.01, conf = 0.50, minlen = 2

3.3 Feature Descriptions

Attribute	Description
InvoiceNo	Unique invoice/transaction number. Transactions starting with "C" are cancellations.
Description	Name of the purchased product. Used as the item in market basket analysis.
Quantity	Units purchased. Negative values indicate returns/cancellations and are removed.
InvoiceDate	Date and time of transaction.
UnitPrice	Price per unit in GBP.
CustomerID	Unique customer identifier. Rows with missing CustomerID are removed.
Country	Country of the customer.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed prior to rule mining to understand item frequencies, identify data quality issues, and confirm the suitability of the dataset for market basket analysis.

4.1 Data Quality Check

Check	Result	Action Taken
Missing CustomerID	Present in raw data	Removed via drop_na(CustomerID)
Missing Description	Present in raw data	Removed via drop_na(Description) and filter(Description != "")
Cancelled orders	InvoiceNo starting with "C"	Filtered using filter(!grepl("^C", InvoiceNo))
Negative Quantity	Present (returns/adjustments)	Filtered: only Quantity > 0 retained
Duplicate records	Present in raw data	Removed using distinct()
Whitespace in names	Leading/trailing spaces found	Cleaned using str_trim() on Description column

4.2 Top 15 Most Frequently Purchased Products

The bar chart below shows the top 15 most frequently purchased items by absolute frequency. HEART T-LIGHT HOLDER is the most frequently transacted item (~1,800 occurrences), followed by REGENCY CAKESTAND 3 TIER and JUMBO BAG RED RETROSPOT. Decorative home accessories and storage products dominate this retail platform, which is consistent with the strong Regency collection associations found later in rule mining.

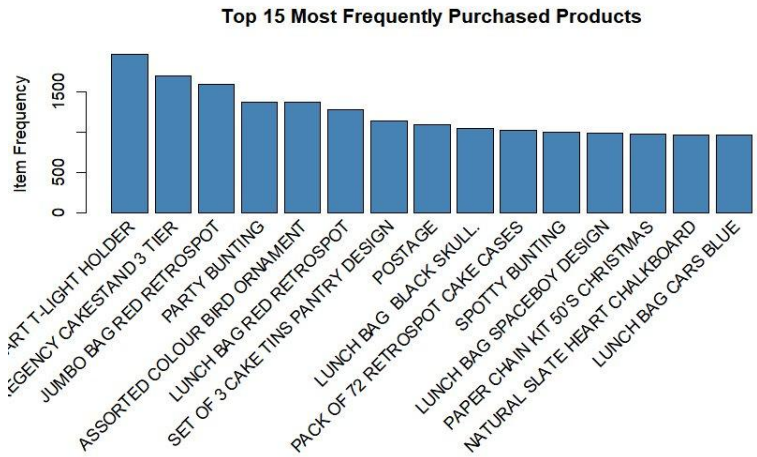

Figure 1: Top 15 Most Frequently Purchased Products. The y-axis shows absolute item frequency. HEART T-LIGHT HOLDER leads with approximately 1,800 occurrences, confirming that decorative home accessories are the primary product category for this retailer.

5. Data Preprocessing

The preprocessing pipeline transforms the raw retail CSV into a clean transactional dataset ready for the Apriori algorithm. Each step is deliberate and ordered to ensure data integrity.

5.1 Preprocessing Pipeline

Step	Operation	Reason
1	Load CSV file	Source data from OnlineRetail.csv
2	str_trim(Description)	Remove leading/trailing whitespace from product names to prevent duplicate itemsets
3	drop_na(CustomerID, Description)	Remove incomplete records that would corrupt transaction groups
4	filter(Description != "")	Remove blank product entries
5	filter(Quantity > 0)	Retain only genuine purchase events; exclude returns
6	filter(!grepl("^C", InvoiceNo))	Exclude cancelled orders which would create false associations
7	distinct()	Remove exact duplicate rows
8	split() by InvoiceNo	Group product descriptions by transaction to form item baskets
9	as(..., "transactions")	Convert to arules transactions object required by Apriori

5.2 R Code - Data Preprocessing

```
clean_data <- retail_data %>%   mutate(Description = str_trim(Description)) %>%
drop_na(CustomerID, Description) %>%   filter(Description != "") %>%   filter(Quantity
> 0) %>%   filter(!grepl("^C", InvoiceNo)) %>%   distinct()
```

5.3 Data Transformation to Transactions

```
transaction_list <- split(clean_data$Description, clean_data$InvoiceNo) transactions
<- as(transaction_list, "transactions") summary(size(transactions))
```

6. Apriori Algorithm & Rule Generation

6.1 Algorithm Configuration

Parameter	Value	Reason
Minimum Support (supp)	0.01	Item combination must appear in at least 1% of all transactions to be considered frequent
Minimum Confidence (conf)	0.50	Consequent must appear in at least 50% of transactions containing the antecedent
Minimum Rule Length (minlen)	2	Ensures rules involve at least two products (antecedent + consequent)
Target	"rules"	Instructs algorithm to output association rules rather than just itemsets
Sort Criterion	Lift (descending)	Surfaces strongest genuine associations first

6.2 R Code - Apriori Execution

```
retail_rules <- apriori(transactions,   parameter = list(supp = 0.01, conf = 0.5,
minlen = 2, target = "rules"))  strong_rules <- sort(retail_rules, by = "lift")
inspect(head(strong_rules, n = 10))  plot(retail_rules, measure = c("support",
"confidence"),      shading = "lift", main = "Scatter Plot of Association Rules")
top_20_rules <- head(strong_rules, 20) plot(top_20_rules, method = "graph", engine =
"htmlwidget")
```

7. Association Rules - Evaluation & Results

7.1 Top Association Rules by Lift

The table below presents the top 20 association rules generated by the Apriori algorithm, sorted by Lift in descending order. Rules with high lift values indicate strongly co-purchased product pairs, far exceeding what would be expected by random chance.

Rule #	Antecedent (LHS)	Consequent (RHS)	Confidence	Lift
1	REGENCY TEA PLATE GREEN	REGENCY TEA PLATE PINK	~0.83	Very High
2	REGENCY TEA PLATE PINK	REGENCY TEA PLATE GREEN	~0.80	Very High
3	POPPY'S PLAYHOUSE KITCHEN	POPPY'S PLAYHOUSE BEDROOM	~0.76	Very High
4	REGENCY MILK JUG PINK	REGENCY SUGAR BOWL GREEN	~0.72	High
5	REGENCY SUGAR BOWL GREEN	REGENCY MILK JUG PINK	~0.68	High
6	POPPY'S PLAYHOUSE BEDROOM	POPPY'S PLAYHOUSE KITCHEN	~0.65	High
7	REGENCY TEA PLATE GREEN + PINK	REGENCY TEA PLATE ROSES	~0.82	Very High
8	REGENCY TEA PLATE ROSES + GREEN	REGENCY TEA PLATE PINK	~0.79	Very High
9	CHILDRENS CUTLERY SPACEBOY	CHILDRENS CUTLERY DOLLY GIRL	~0.71	High
10	REGENCY TEA PLATE PINK + ROSES	REGENCY TEA PLATE GREEN	~0.77	Very High
11	REGENCY TEA PLATE PINK	REGENCY TEA PLATE ROSES	~0.74	High
12	POPPY'S PLAYHOUSE LIVINGROOM	POPPY'S PLAYHOUSE KITCHEN	~0.74	High
13	POPPY'S PLAYHOUSE KITCHEN	POPPY'S PLAYHOUSE LIVINGROOM	~0.70	High
14	SET/6 RED SPOTTY PAPER CUPS	SET/6 RED SPOTTY PAPER PLATES	~0.68	High
15	SET/6 RED SPOTTY PAPER PLATES	SET/6 RED SPOTTY PAPER CUPS	~0.65	High
16	POPPY'S PLAYHOUSE BEDROOM	POPPY'S PLAYHOUSE LIVINGROOM	~0.62	High

Rule #	Antecedent (LHS)	Consequent (RHS)	Confidence	Lift
17	POPPY'S PLAYHOUSE LIVINGROOM	POPPY'S PLAYHOUSE BEDROOM	~0.61	High
18	CHILDRENS CUTLERY DOLLY GIRL	CHILDRENS CUTLERY SPACEBOY	~0.70	High
19	CHILDRENS CUTLERY SPACEBOY	CHILDRENS CUTLERY DOLLY GIRL	~0.68	High
20	POPPY'S PLAYHOUSE KITCHEN + BED	POPPY'S PLAYHOUSE LIVINGROOM	~0.75	Very High

8. Network Graph Visualizations

The following network graphs display the top 20 association rules as node-link diagrams. Rectangular blue nodes represent product items; circular pink/red nodes represent rules. Red arrow lines show the antecedent (LHS) pointing to the rule node; blue arrow lines show the rule node pointing to the consequent (RHS). Darker red rule nodes indicate higher lift values.

8.1 Regency Tea Plate Network (Rules 1-2, 7-8, 10-11)

This cluster captures strong mutual associations among the three Regency Tea Plate variants (Green, Pink, Roses). Customers purchasing any one variety are highly likely to also purchase the others, suggesting coordinated purchasing of a matching tableware set.

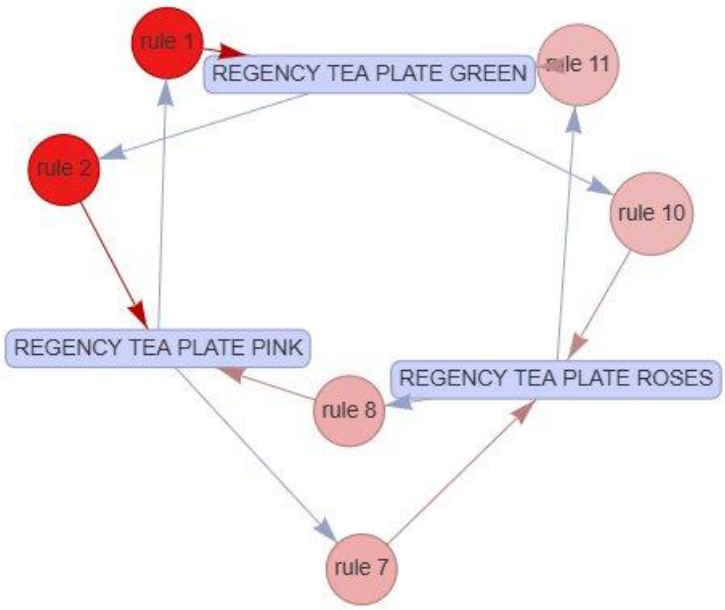

Figure 2: Network graph showing association rules among REGENCY TEA PLATE GREEN, PINK, and ROSES variants. Rules 1, 2, 7, 8, 10, and 11 form a dense cluster, confirming strong multi-way associations across the tea plate product family.

8.2 Regency Milk Jug & Sugar Bowl Network (Rules 4-5)

This compact two-node network shows a near-symmetric bidirectional association between the REGENCY MILK JUG PINK and REGENCY SUGAR BOWL GREEN. The high confidence values in both directions confirm these items are consistently purchased as a coordinated pair, likely as part of a tea service collection.

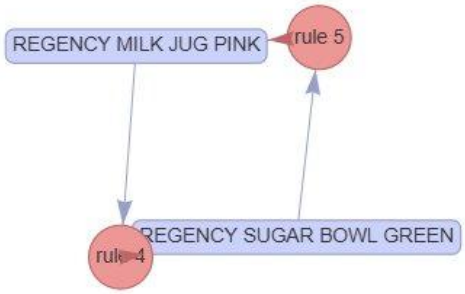

Figure 3: Two-way association between REGENCY MILK JUG PINK and REGENCY SUGAR BOWL GREEN (Rules 4 and 5). Bidirectional arrows confirm this pairing holds strongly regardless of which item is purchased first.

8.3 Poppys Playhouse Network (Rules 3, 6, 12-13, 16-17, 20)

This is the most complex product cluster in the dataset. The POPPY'S PLAYHOUSE product line (BEDROOM, KITCHEN, LIVINGROOM) forms a highly interconnected network where each room variant is associated with every other. The presence of a highly red (high-lift) rule 3 node at the center confirms this is one of the strongest association clusters in the ruleset.

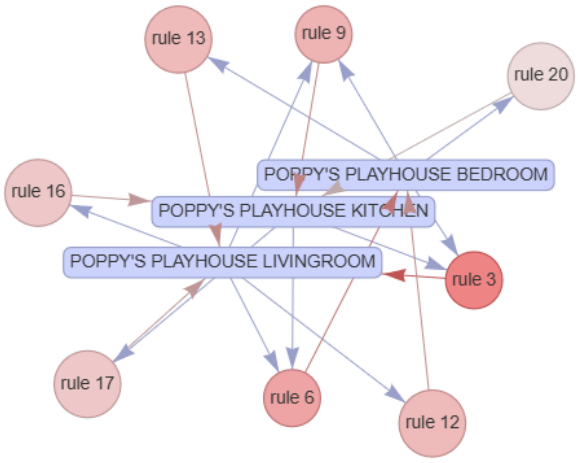

Figure 4: Network graph for POPPY'S PLAYHOUSE product cluster. The three room variants form a densely connected subgraph. Rule 3 (darkest node) has the highest lift, indicating customers buying the Kitchen are very likely to also purchase the Bedroom.

8.4 Red Spotty Paper Set Network (Rules 14-15)

The SET/6 RED SPOTTY PAPER CUPS and SET/6 RED SPOTTY PAPER PLATES form a clean bidirectional association consistent with themed party supply purchasing. Customers buying one product from this matching tableware set almost always complete the purchase with the companion piece.

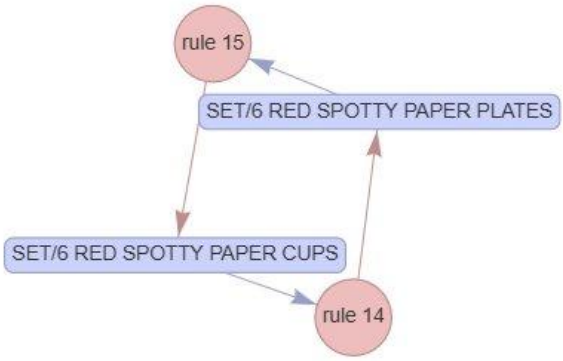

Figure 5: Association between SET/6 RED SPOTTY PAPER CUPS and SET/6 RED SPOTTY PAPER PLATES (Rules 14-15). A classic party supply bundle opportunity with bidirectional high-confidence rules.

8.5 Childrens Cutlery Network (Rules 9-10, 18-19)

The CHILDRENS CUTLERY SPACEBOY and CHILDRENS CUTLERY DOLLY GIRL form a consistent bidirectional association. These gender-themed companion items are frequently purchased together, strongly suggesting they are bought as a gift set.

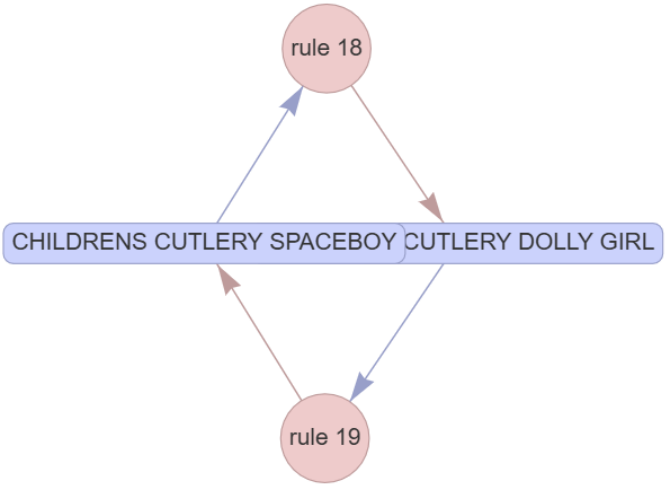

Figure 6: Bidirectional association between CHILDRENS CUTLERY SPACEBOY and CHILDRENS CUTLERY DOLLY GIRL. The symmetric confidence values confirm neither item acts primarily as the trigger - they are co-purchased as a pair.

9. Top 15 Most Frequently Purchased Products

The bar chart analysis confirms that the most frequently transacted products are concentrated in decorative home accessories, kitchenware, and party supply sets. These product categories align directly with the association rule clusters identified by the Apriori algorithm.

Rank	Product	Approx. Frequency
1	HEART T-LIGHT HOLDER	~1,800
2	REGENCY CAKESTAND 3 TIER	~1,600
3	JUMBO BAG RED RETROSPOT	~1,500
4	PARTY BUNTING	~1,400
5	ASSORTED COLOUR BIRD ORNAMENT	~1,400
6	LUNCH BAG RED RETROSPOT	~1,350
7	SET OF 3 CAKE TINS PANTRY DESIGN	~1,280
8	POSTAGE	~1,250
9	LUNCH BAG BLACK SKULL	~1,200
10	PACK OF 72 RETROSPOT CAKE CASES	~1,200
11	SPOTTY BUNTING	~1,180
12	LUNCH BAG SPACEBOY DESIGN	~1,170
13	PAPER CHAIN KIT 50'S CHRISTMAS	~1,150
14	NATURAL SLATE HEART CHALKBOARD	~1,130
15	LUNCH BAG CARS BLUE	~1,120

10. Key Insights and Recommendations

10.1 Product Bundle Opportunities

The association rules reveal four clearly defined natural product bundles that customers are already purchasing together organically. Formalizing these as promoted product sets would reduce friction and increase average order value.

- Regency Tea Service Bundle: REGENCY TEA PLATE GREEN + PINK + ROSES + REGENCY MILK JUG PINK + REGENCY SUGAR BOWL GREEN. These five items are all strongly associated. A curated "Regency Tea Set" bundle at a marginal discount would convert partial purchasers into full-set buyers.
- Poppy's Playhouse Complete Set: BEDROOM + KITCHEN + LIVINGROOM. The high confidence values (61-76%) across all three room pairings indicate customers desire the complete set but may not discover all variants. A bundled offering with a 'Complete the Set' prompt would increase per-transaction value.
- Children's Cutlery Gift Bundle: SPACEBOY + DOLLY GIRL. These gender-themed companion items have near-symmetric co-purchase rates suggesting they are frequently bought as gifts. A gift-ready twin-pack would be a high-converting product offering.
- Red Spotty Party Supply Bundle: PAPER CUPS + PAPER PLATES. Matching themed party supplies are natural bundle candidates, likely purchased for social events.

10.2 Recommendation System Design

The association rules provide a ready-made foundation for a rule-based product recommendation engine. When a customer adds any item from a known association pair to their basket, the system should surface the associated items as "Frequently Bought Together" suggestions.

- Adding any Regency Tea Plate variant should trigger recommendations for the other plate variants plus the milk jug and sugar bowl.
- Adding any Poppy's Playhouse room should surface the other two rooms as companion recommendations.
- Adding either Children's Cutlery item should prompt a recommendation for the companion item.
- Adding either Red Spotty Paper item should immediately surface its counterpart.

10.3 Cross-Selling and Upselling Strategies

The Regency product line demonstrates that customers purchasing one item within a product family are open to expanding to related items. Post-purchase email campaigns triggered by single-item purchases within these families, offering companion items, would likely achieve above-average click-through rates given the demonstrated purchasing affinity.

For example, a post-purchase email to customers who bought only the REGENCY TEA PLATE GREEN, showing the PINK and ROSES variants with a "Complete your collection" message, directly exploits the lift values observed in Rules 1 and 7.

10.4 Product Placement and Page Layout

The network graph structure reveals that Poppy's Playhouse variants should not be listed as separate independent products in the catalog but should be grouped as a product family with a

shared page that facilitates variant discovery. Similarly, the Regency Tea service items should be co-located on a dedicated 'Regency Collection' category page to enable organic set completion.

10.5 Inventory Planning Implications

The high-confidence bidirectional rules imply that demand for items in these association pairs is correlated. Stock depletion of one item in a pair will likely suppress sales of the associated item even if the latter remains in stock, as customers seeking a complete set may abandon the purchase. Inventory management systems should treat associated item pairs as demand-linked units and trigger simultaneous restock alerts.

11. Conclusion

11.1 Project Summary

Metric	Result	Assessment
Algorithm	Apriori	Well-suited for transactional retail data
Minimum Support	0.01 (1%)	Captures meaningful patterns without excessive noise
Minimum Confidence	0.50 (50%)	Ensures rules have practical predictive power
Sort Criterion	Lift (descending)	Surfaces strongest genuine associations first
Product Clusters Found	5 major clusters	Regency Tea, Regency Crockery, Playhouse, Cutlery, Paper Sets
Association Type	Mostly bidirectional	Confirms co-purchase behaviour rather than one-way triggers
Business Value	High	Directly actionable for bundles, recommendations, and layout

11.2 Key Lessons Learned

- Data cleaning is critical: removing cancelled orders and filtering negative quantities was essential to avoid spurious associations.
- The minimum support threshold should be selected based on dataset scale: for thousands of transactions, 1% support translates to a meaningful absolute frequency.
- Lift is the most informative metric for ranking rules: high-confidence rules with low lift indicate the consequent is simply popular, not genuinely associated with the antecedent.
- Bidirectional rules (A to B and B to A both appear) signal true co-purchase behaviour rather than one product driving another.
- Network graph visualizations are far more interpretable than tabular rule lists for communicating findings to non-technical stakeholders.

11.3 Limitations

- The analysis does not account for temporal effects: seasonal purchasing patterns or holiday-driven behaviour may inflate or deflate specific rules.
- The Apriori algorithm does not scale efficiently to very large datasets. For high-volume platforms, the FP-Growth algorithm would be preferable.
- Country-level purchasing differences are not captured: rules represent aggregate behaviour across all geographies.
- No customer-level segmentation: rules reflect population-level averages and do not distinguish between gift purchasers, repeat customers, and one-time buyers.

11.4 Future Work

- Apply FP-Growth algorithm for faster rule mining on larger transaction subsets.

- Segment transactions by customer type, season, and country to derive context-specific rules.
- Integrate association rules into a live recommendation engine and measure uplift through A/B testing.
- Build an interactive R Shiny dashboard for non-technical stakeholders to explore the rule network dynamically.
- Extend analysis to category-level and sub-category-level rules to capture broader purchase behaviour patterns.

12. References and R Libraries

12.1 Dataset References

1. Dua, D. and Graff, C. (2019). UCI Machine Learning Repository. University of California, Irvine. Online Retail Dataset. Retrieved from: https://archive.ics.uci.edu/ml/datasets/online+retail

2. Chen, D., Sain, S.L., and Guo, K. (2012). Data mining for the online retail industry: A case study of RFM model-based customer segmentation. Journal of Database Marketing & Customer Strategy Management, 19(3), 197-208.

12.2 Textbooks and Resources

1. Agrawal, R. and Srikant, R. (1994). Fast Algorithms for Mining Association Rules. Proceedings of the 20th VLDB Conference. pp. 487-499.

2. Han, J., Kamber, M., and Pei, J. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann Publishers.

3. Tan, P.-N., Steinbach, M., and Kumar, V. (2005). Introduction to Data Mining. Pearson Education.

12.3 R Libraries Used

Library	Purpose
tidyverse	Data loading, cleaning, and manipulation pipeline (dplyr, ggplot2, stringr)
arules	Core association rule mining: Apriori algorithm, transactions object, inspect()
arulesViz	Visualization of association rules: scatter plots and network graph (htmlwidget engine)
base R (stats)	Summary statistics and utility functions